Pushpraj Singh Khichi

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.89
SHARDA VIDYA MANDIR	Class XII (CBSE)	2022	94.60%
SHARDA VIDYA MANDIR	Class X (CBSE)	2020	93.40%

WORK EXPERIENCE

Software Development Intern- Kotak Mahindra Bank

- Developed backend services and REST APIs using Java and Spring Boot, with MySQL, Hibernate, Maven, and Docker for scalable application development and reliable deployment across environments.
- Contributed to the **NOC Automation Engine**, optimizing certificate-issuance workflows using **Temporal** and enabling parallel API execution to improve overall system latency, throughput, and operational efficiency.
- Implemented a **multi-channel communication system** and reduced redundant API calls, improving workflow scalability, system reliability, and overall service performance.
- Built **Grafana and Prometheus monitoring dashboards** to track latency, performance, and system health, improving application observability, troubleshooting, and real-time debugging capabilities.
- Implemented **error classification** and explored **PagerDuty** for incident alerting and scheduling, supporting proactive issue detection, faster troubleshooting, and effective incident management.
- Collaborated with the development team across **local, development, and production environments**, following Git-based workflows and agile software development practices for collaborative feature delivery.

PROJECTS

Distributed File System with Caching Mechanism

- Designed and implemented a distributed file storage system using **TCP socket communication**, with a Naming Server coordinating multiple Storage Servers for file operations and metadata management.
- Implemented **LRU-based in-memory caching** and efficient file-path indexing using hashing/data structures to reduce repeated lookups and improve overall file retrieval latency across storage servers.
- Added **multithreaded request handling, concurrency control, and replica synchronization** to support multiple concurrent client requests and improve data availability and fault tolerance.

Personal Finance Tracker

- Developed a full-stack finance management application to track **category-wise expenses, multiple income sources,** and configurable spending limits, enabling users to monitor and manage their financial activities.
- Implemented **JWT-based authentication** with secure user registration, login, and user-specific financial data.
- Built **Spring Boot REST APIs** for expense, income, category, and budget-limit management.
- Developed and integrated the frontend with backend APIs for a complete personal finance management experience.

Resume-Job Matching System

- Developed a **RAG-based job recommendation system** using dense vector retrieval and semantic similarity to match candidate resumes against a structured job corpus based on skills, experience, and job requirements.
- Implemented **ChromaDB with HNSW-based** approximate nearest-neighbor search for **top-K candidate retrieval**, followed by cross-encoder reranking to perform contextual relevance scoring and improve matching precision.
- Built an end-to-end retrieval and generation pipeline with resume preprocessing, embedding generation, metadata filtering, semantic search, candidate reranking, and LLM-based grounded job-match explanations.

Metro Route Finder

- Developed a **Metro Route Finder** in C++ to determine the **shortest route** between any two metro stations using Dijkstra's shortest-path algorithm with weighted graph representation and STL-based data structures.
- Modeled the metro network as a **weighted graph**, with stations as vertices and travel distances as edge weights.
- Implemented **STL** including vector, map, and priority queue for efficient graph representation and traversal.
- Built an interface to accept **source and destination stations** and display the optimal route, distance, and fare.

SKILLS

Programming Languages: C++, Java, Python, SQL

CS Fundamentals: Data Structures and Algorithms (DSA), Object-Oriented Programming (OOP), Database Management

Systems (DBMS), Operating Systems (OS), Computer Networks (CN), System Design, Classical Machine Learning

Tools: VS Code, Intellij, Docker, MYSQL, Postman, Azure Devops

SCHOLASTIC ACHIEVEMENTS

- **Solved 1000+** DSA problems across coding platforms, including **LeetCode, Codeforces, and GeeksforGeeks.**

RELEVANT COURSES

Mathematics: Calculus 1, Calculus 2, Elementary Linear Algebra, Differential Equations, Transform Techniques, Probability, Complex Variables, Statistics.

Programming: Introduction to Programming, Data Structures and Algorithms.

Extra Courses: Spoken Japanese, Communication Skills, Science-Technology and Society, Introduction to Health Economics, Fundamentals of Organizational Structure.

EXTRACURRICULAR

- **Team Captain, 3rd Place - MOM (Mad Over Marketing)**, E-Cell, IIT Hyderabad.
- Represented department in **Inter-Department Cricket** competitions as a team representative.
- Represented institute teams in **Inter-Hostel & Inter-Department Basketball** competitions at institute-level events.